

Class 19: Pertussis Mini Project

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Background

Pertussis is a bacterial lung infection also known as Whooping cough. Let's begin by examining CDC reported case numbers in the US:

```
cdc <- data.frame(
  year = c(1922L, 1923L, 1924L, 1925L,
           1926L, 1927L, 1928L, 1929L, 1930L, 1931L,
           1932L, 1933L, 1934L, 1935L, 1936L,
           1937L, 1938L, 1939L, 1940L, 1941L, 1942L,
           1943L, 1944L, 1945L, 1946L, 1947L,
           1948L, 1949L, 1950L, 1951L, 1952L,
           1953L, 1954L, 1955L, 1956L, 1957L, 1958L,
           1959L, 1960L, 1961L, 1962L, 1963L,
           1964L, 1965L, 1966L, 1967L, 1968L, 1969L,
           1970L, 1971L, 1972L, 1973L, 1974L,
           1975L, 1976L, 1977L, 1978L, 1979L, 1980L,
           1981L, 1982L, 1983L, 1984L, 1985L,
           1986L, 1987L, 1988L, 1989L, 1990L,
           1991L, 1992L, 1993L, 1994L, 1995L, 1996L,
           1997L, 1998L, 1999L, 2000L, 2001L,
           2002L, 2003L, 2004L, 2005L, 2006L, 2007L,
           2008L, 2009L, 2010L, 2011L, 2012L,
           2013L, 2014L, 2015L, 2016L, 2017L, 2018L,
           2019L, 2020L, 2021L, 2022L, 2023L, 2024L),
  cases = c(107473, 164191, 165418, 152003,
            202210, 181411, 161799, 197371,
            166914, 172559, 215343, 179135, 265269,
            180518, 147237, 214652, 227319, 103188,
            183866, 222202, 191383, 191890, 109873,
            133792, 109860, 156517, 74715, 69479,
            120718, 68687, 45030, 37129, 60886,
```

```

        62786,31732,28295,32148,40005,
        14809,11468,17749,17135,13005,6799,
        7717,9718,4810,3285,4249,3036,
        3287,1759,2402,1738,1010,2177,2063,
        1623,1730,1248,1895,2463,2276,
        3589,4195,2823,3450,4157,4570,
        2719,4083,6586,4617,5137,7796,6564,
        7405,7298,7867,7580,9771,11647,
        25827,25616,15632,10454,13278,
        16858,27550,18719,48277,28639,32971,
        20762,17972,18975,15609,18617,
        6124,2116,3044,7063, 22538)
    )

```

Plot of cases per year for Pertussis in the US

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

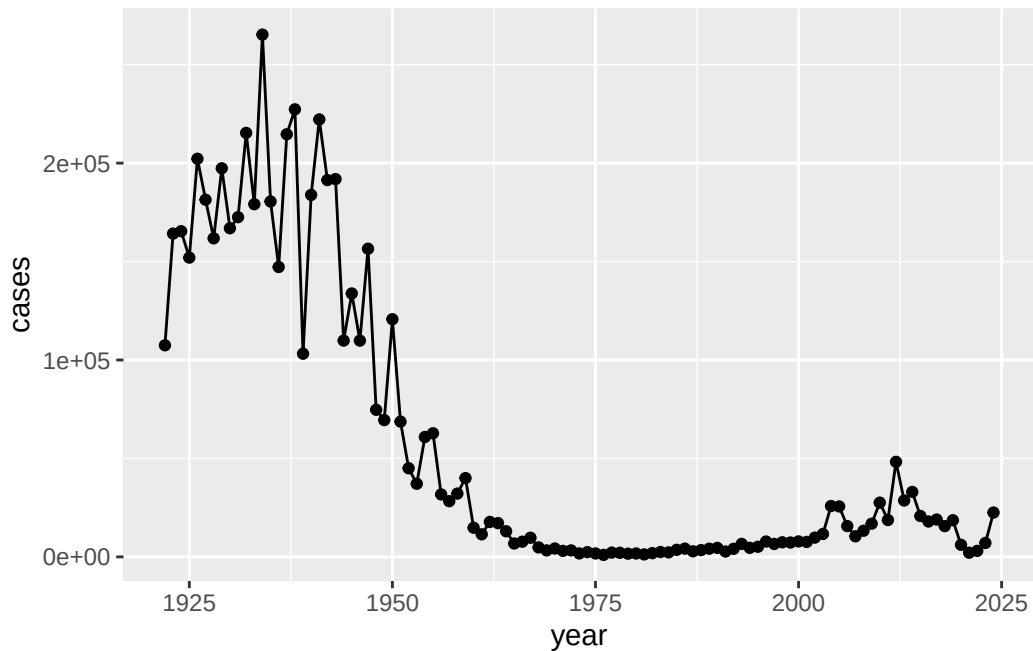
```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.5.2

```

ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line()

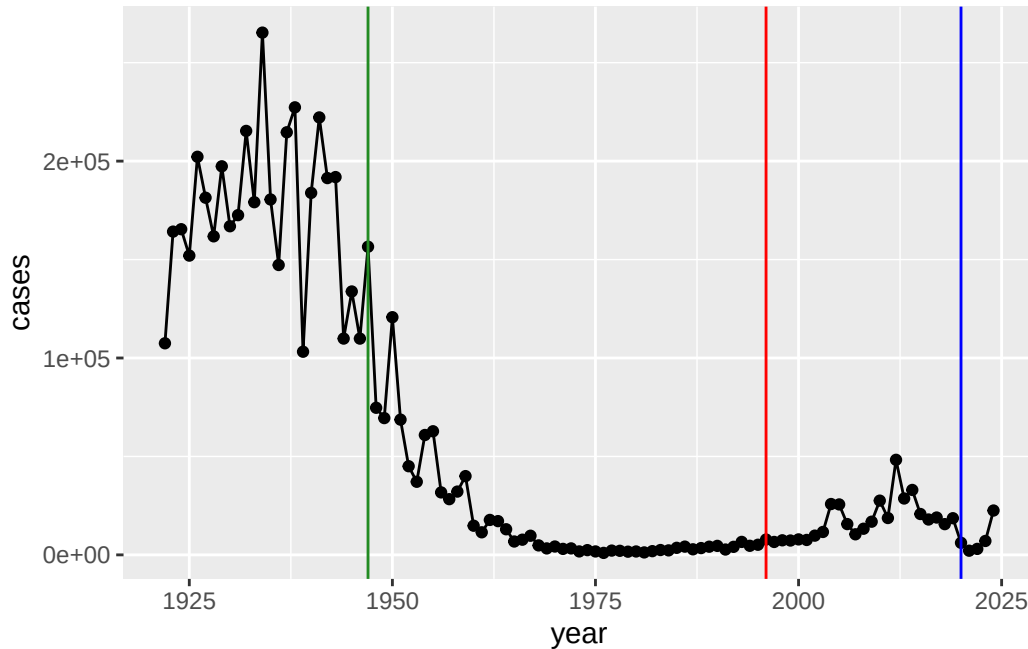
```



Add some major milestone timepoints to our plot:

Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 1947, col="forestgreen") +
  geom_vline(xintercept = 1996, col="red") +
  geom_vline(xintercept = 2020, col="blue")
```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend? The full introduction of the mandatory wP (whole-cell) Pertussis immunization in the mid 1940s led to a dramatic reduction in case numbers (from over 200,000 to 100s).

The switch to the aP (newer acellular formalization) had a lag period where people were switching from the wP to the aP vaccine that was followed by a large increase in case numbers with a peak of nearly 50,000 cases.

The 2020 lock-downs and social distancing measures showed a noticeable decrease in case numbers that slowly increased again once lockdown and social distancing was unregulated.

The CMI-PB project

The mission of CMI-PB is to provide the scientific community with a comprehensive, high-quality and freely accessible resource of Pertussis booster vaccination.

Website: <https://www.cmi-pb.org>

They make their data available via JSON format API endpoints - basically the database tables in a key:value type format like "infancy_vac": "wP". To read this, we can use the `read_json()` function from the `jsonlite` package. Install with `install.packages("jsonlite")`

```
library(jsonlite)
```

Warning: package 'jsonlite' was built under R version 4.5.2

```
subject <- read_json(path="http://cmi-pb.org/api/v5_1/subject",
                      simplifyVector=TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female Not	Hispanic or Latino	White
2	2	wP	Female Not	Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male Not	Hispanic or Latino	Asian
5	5	wP	Male Not	Hispanic or Latino	Asian
6	6	wP	Female Not	Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

Q. How many “subjects”/individuals are in this dataset?

```
nrow(subject)
```

```
[1] 172
```

Q4. How many aP and wP subjects are there?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. What is the breakdown by “biological_sex” and “race”?

```
table(subject$biological_sex)
```

```
Female    Male
    112     60
```

```
table(subject$race)
```

```

American Indian/Alaska Native
                                1
                                Asian
                                44
Black or African American
                                5
More Than One Race
                                19
Native Hawaiian or Other Pacific Islander
                                2
Unknown or Not Reported
                                21
White
                                80
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

This breakdown is not particularly representative of the US population - this is a serious caveat for this study. However, it is still the largest sample of its type ever assembled.

```
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```
head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost	
1	1	1	-3	
2	2	1	1	
3	3	1	3	
4	4	1	7	
5	5	1	11	
6	6	1	32	

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
subject$dob <- ymd(subject$year_of_birth)
subject$age <- today() - subject$dob
```

```
ap <- subject %>% filter(infancy_vac == "aP")
round(summary(time_length(ap$age, "years")))
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	27	28	28	29	35

```
wp <- subject %>% filter(infancy_vac == "wP")
round( summary( time_length( wp$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
23	33	35	37	40	58

Q8. Determine the age of all individuals at time of boost?

```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

We need to “join” or link these tables with the `subject` table so we can begin to analyze this data and know who a given Ab sample was collected for and when.

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:


```
library(dplyr)
```

```
meta <- inner_join(subject, specimen)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	subject_id	infancy_vac	biological_sex		ethnicity	race	
1	1	wP	Female	Not Hispanic or Latino	White		
2	1	wP	Female	Not Hispanic or Latino	White		
3	1	wP	Female	Not Hispanic or Latino	White		
4	1	wP	Female	Not Hispanic or Latino	White		
5	1	wP	Female	Not Hispanic or Latino	White		
6	1	wP	Female	Not Hispanic or Latino	White		

	year_of_birth	date_of_boost	dataset	dob	age	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	1
2	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	2
3	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	3
4	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	4
5	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	5
6	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	6

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3	0	Blood
2	1	1	Blood
3	3	3	Blood
4	7	7	Blood
5	11	14	Blood
6	32	30	Blood

	visit
1	1
2	2
3	3
4	4
5	5
6	6

Now let's join the `ab_titer` table with our `meta` table so we have all the information about a given Ab measurement

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
ab_data <- inner_join(meta, ab_titer)
```

Joining with `by = join\_by(specimen\_id)`

```
head(ab_data)
```

	subject_id	infancy_vac	biological_sex		ethnicity	race	
1	1	wP	Female	Not Hispanic or Latino	White		
2	1	wP	Female	Not Hispanic or Latino	White		
3	1	wP	Female	Not Hispanic or Latino	White		
4	1	wP	Female	Not Hispanic or Latino	White		
5	1	wP	Female	Not Hispanic or Latino	White		
6	1	wP	Female	Not Hispanic or Latino	White		

	year_of_birth	date_of_boost	dataset	dob	age	specimen_id
1	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	1
2	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	1
3	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	1
4	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	1
5	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	1
6	1986-01-01	2016-09-12	2020_dataset	1986-01-01	14585 days	1

	actual_day_relative_to_boost	planned_day_relative_to_boost	specimen_type
1	-3		Blood
2	-3		Blood
3	-3		Blood
4	-3		Blood
5	-3		Blood
6	-3		Blood

	visit	isotype	is_antigen_specific	antigen	MFI	MFI_normalised	unit
1	1	IgE	FALSE	Total	1110.21154	2.493425	UG/ML
2	1	IgE	FALSE	Total	2708.91616	2.493425	IU/ML
3	1	IgG	TRUE	PT	68.56614	3.736992	IU/ML
4	1	IgG	TRUE	PRN	332.12718	2.602350	IU/ML
5	1	IgG	TRUE	FHA	1887.12263	34.050956	IU/ML
6	1	IgE	TRUE	ACT	0.10000	1.000000	IU/ML

	lower_limit_of_detection
1	2.096133
2	29.170000
3	0.530000

```
4          6.205949
5          4.679535
6          2.816431
```

Q. How many Ab (antibodies) measurements do we have in total?

```
nrow(ab_data)
```

```
[1] 61956
```

Q. How many different isotypes (types of Ab) are in the dataset?

```
unique(ab_data$isotype)
```

```
[1] "IgE" "IgG" "IgG1" "IgG2" "IgG3" "IgG4"
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(ab_data$isotype)
```

```
 IgE  IgG  IgG1  IgG2  IgG3  IgG4
6698 7265 11993 12000 12000 12000
```

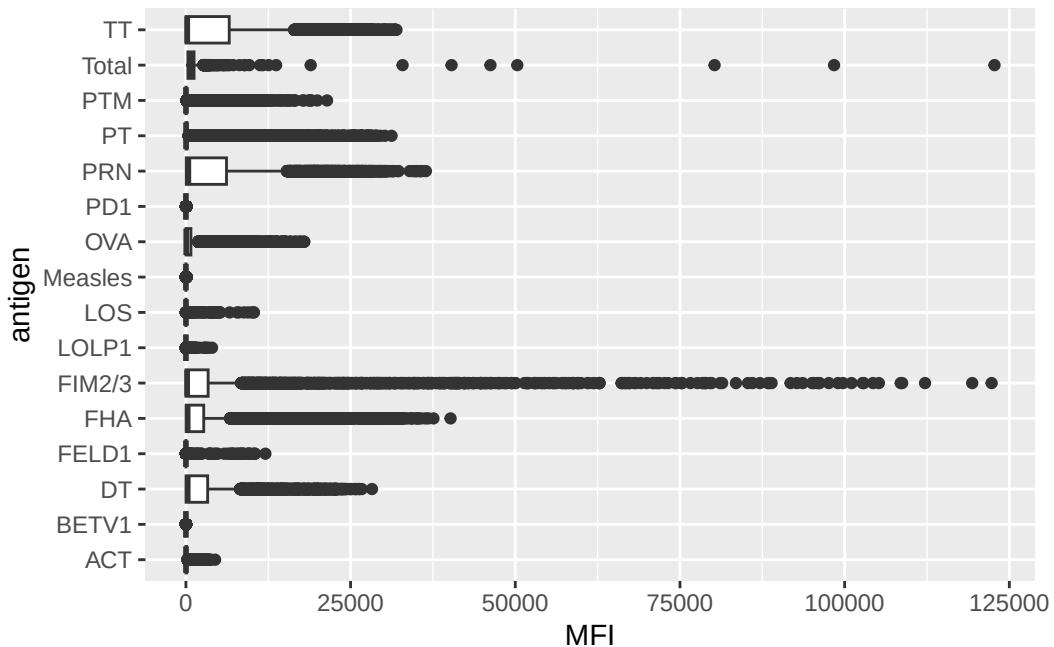
Q. How many different antigens?

```
unique(ab_data$antigen)
```

```
[1] "Total" "PT" "PRN" "FHA" "ACT" "LOS" "FELD1"
[8] "BETV1" "LOLP1" "Measles" "PTM" "FIM2/3" "TT" "DT"
[15] "OVA" "PD1"
```

```
ggplot(ab_data) +
  aes(MFI, antigen) +
  geom_boxplot()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_boxplot()`).
```



Q12. What are the different `$dataset` values in `abdata` and what do you notice about the number of rows for the most “recent” dataset?

```
table(ab_data$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520       8085       7301      15050
```

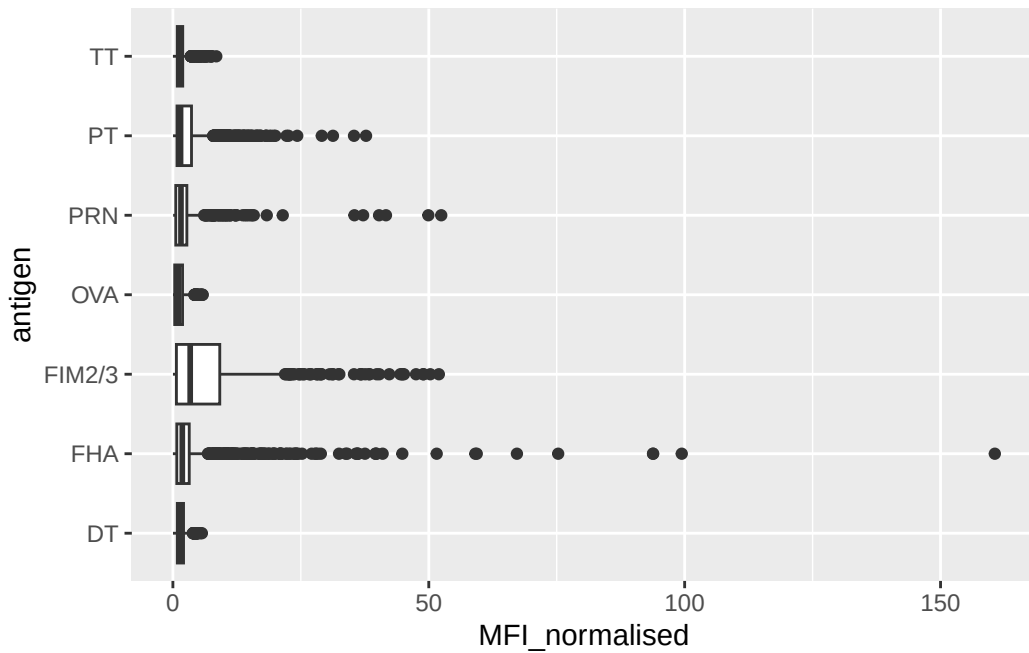
The most “recent” dataset has a much larger sample size compared to the previous years.

Examine IgG Ab titer levels

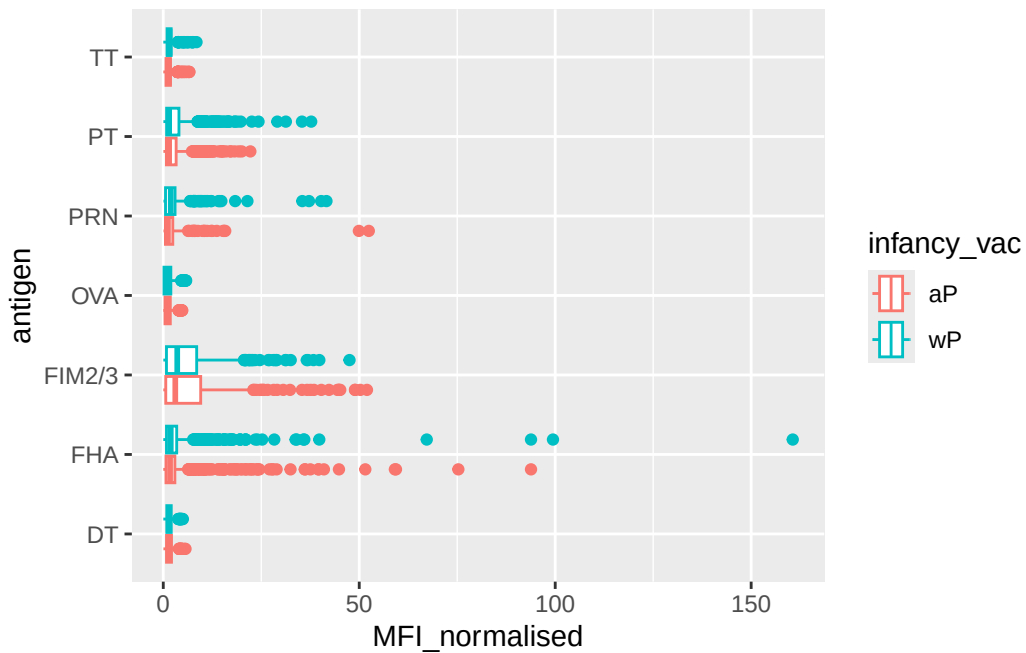
IgG is crucial for long-term immunity and responding to bacterial & viral infections.

```
igg <- ab_data |>
  filter(isotype=="IgG")
```

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()
```

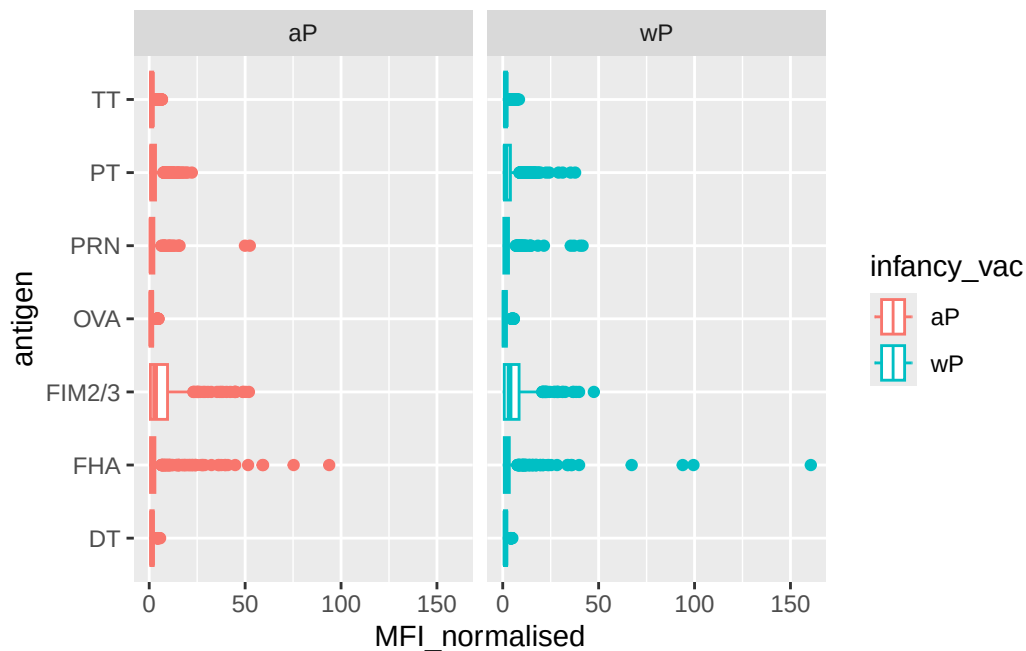


```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



We can “facet” our plot by wP vs aP

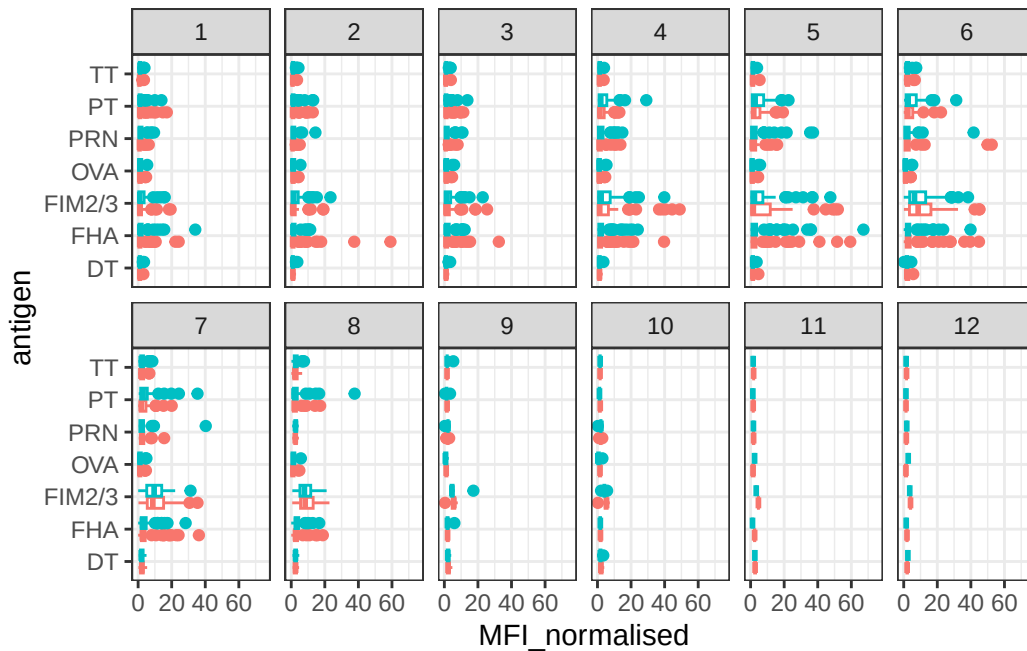
```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow=2) +
  xlim(0,75) +
  theme_bw()
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

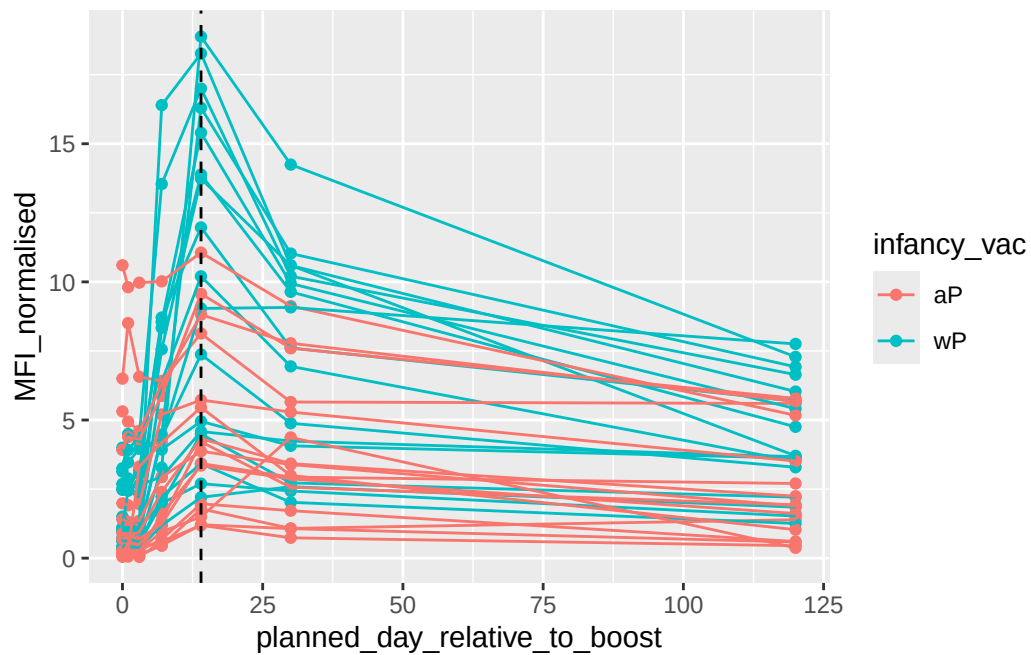


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

PT, PRN, and FHA because for the aP immunization, the effectiveness may decrease over time as shown by the decreasing IgG antibody levels. Others may not show these differences because these vaccinations might not have these antigens.

More advanced analysis digging into individual antigen responses over time:

```
filter(igg, antigen == "PT", dataset == "2021_dataset") |>
  ggplot() +
  aes(x=planned_day_relative_to_boost,
      y=MFI_normalised,
      col=infancy_vac,
      group=subject_id) +
  geom_point() +
  geom_line() +
  geom_vline(xintercept=14, linetype="dashed")
```



This plot shows the time course of Pertussis toxin (PT) antibody responses for a large set of wP (teal color) and aP (red color) individuals. Levels peak at day 14 and are larger in magnitude for wP than aP individuals.

There are lots of cool things to explore in this dataset and we need coding and biology knowledge to do it effectively.